

Software Security Analysis

The Internals of a Windows System Using Different Tools

Introduction

This lab is designed to work with Microsoft Sysinternals tools Process Monitor (*procmon*) and *Autoruns* to study about the internal operation of a windows system. These tools provide detailed insight into how applications interact with the Windows operating system at the file system, process, and registry levels. Notepad was used as a sample application to monitor how a simple program accesses and interacts with system resources using Process Monitor. Autoruns tool is used to analyze startup entries and identify programs configured to execute automatically during system boot or user logon.

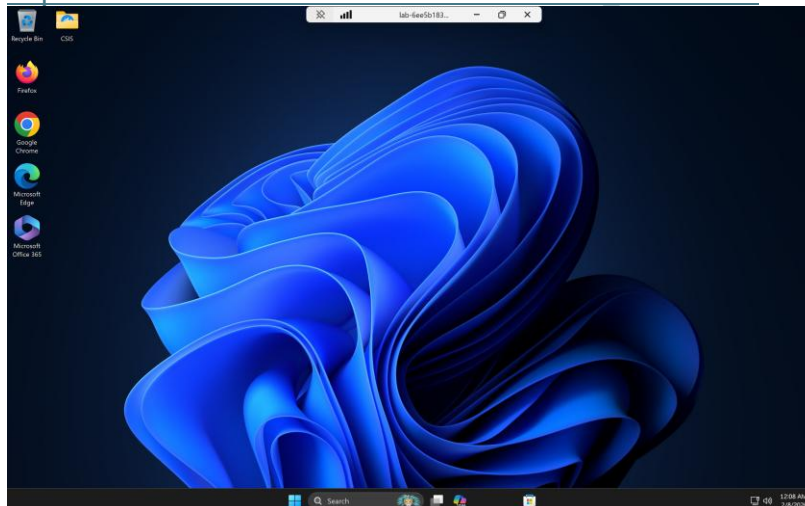
Process monitor can capture real-time system activity, including registry operations, file system access, process activity, and thread events. By using different filters, specific details like registry queries or file access events can be isolated and monitored. The Autoruns can provide details of startup-related entries, including those located in registry run keys, Active Setup components, services, and scheduled tasks.

Having a baseline of normal system behavior allows analysts to compare standard operating patterns with the detailed provided by these tools. These details can be used during investigations to detect suspicious modifications, unauthorized startup entries, and unauthorized activities in the system.

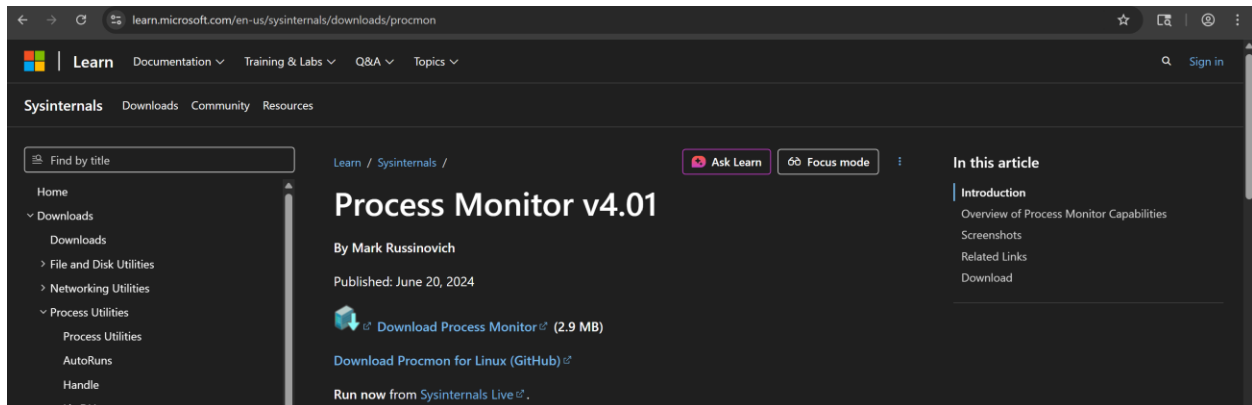
Part 1 - Process Monitor (procmon)

1. Logged to Azure Lab environment using following web link and logged into the Windows VM using Oracle VirtualBox.

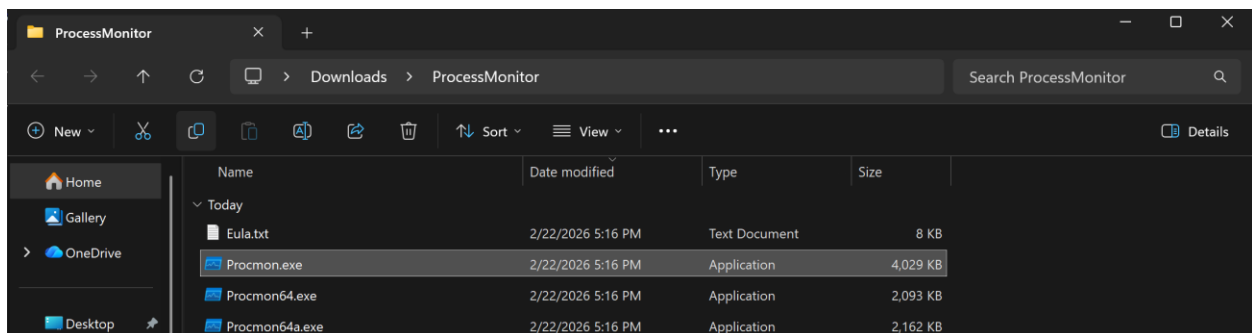
https://labs.azure.com/virtualmachines?feature_vnext=true



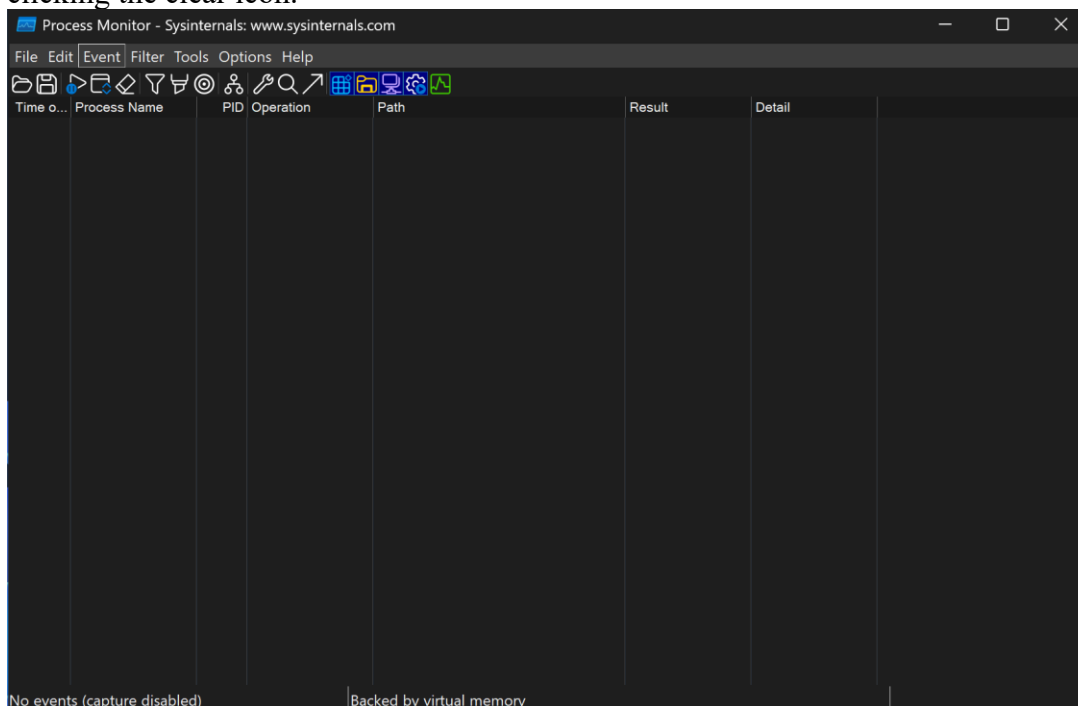
2. Used URL: <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon> and downloaded the process Monitor.



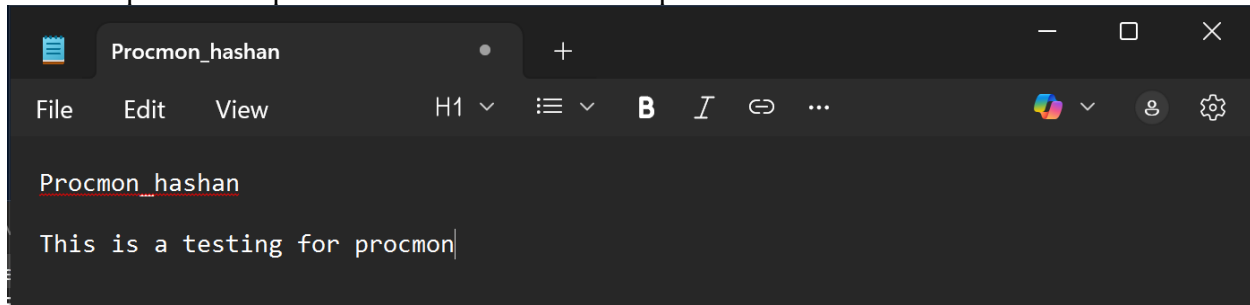
3. Extracted the downloaded file and ran Procmon.exe to install procmon.



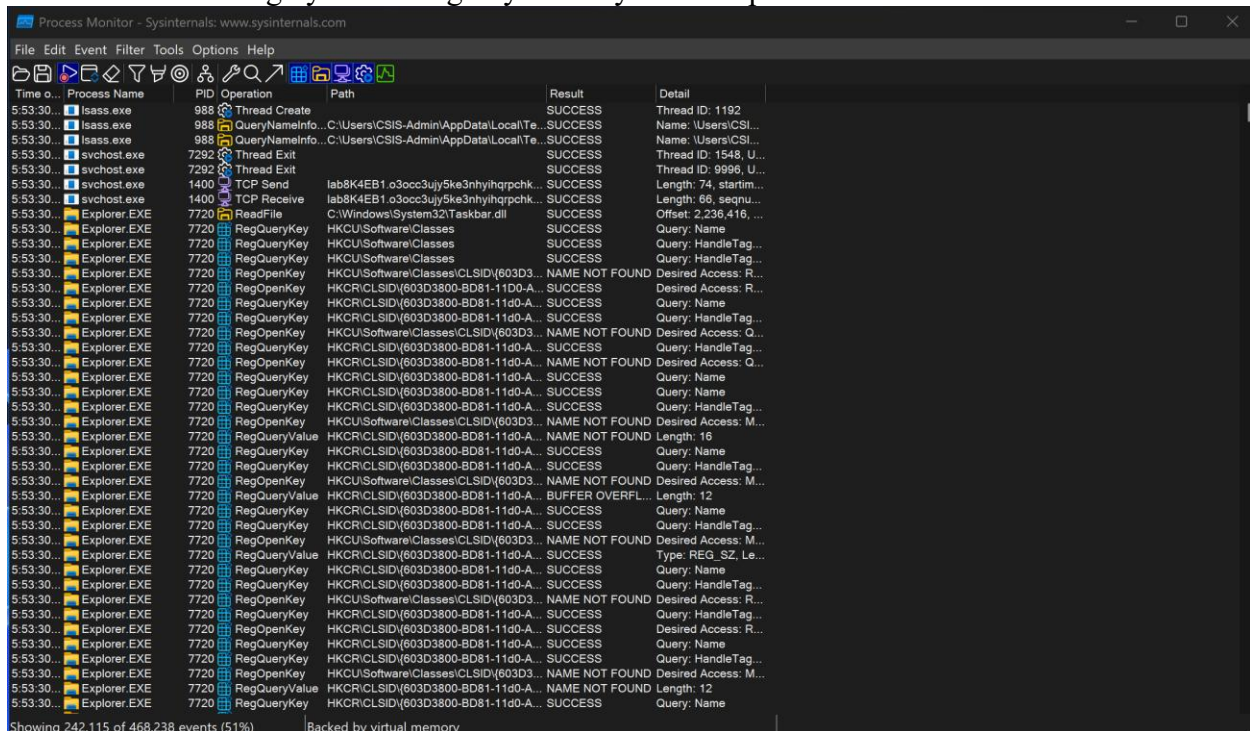
4. Stopped capture in procmon by clicking the capture icon and then clear capture by clicking the clear icon.



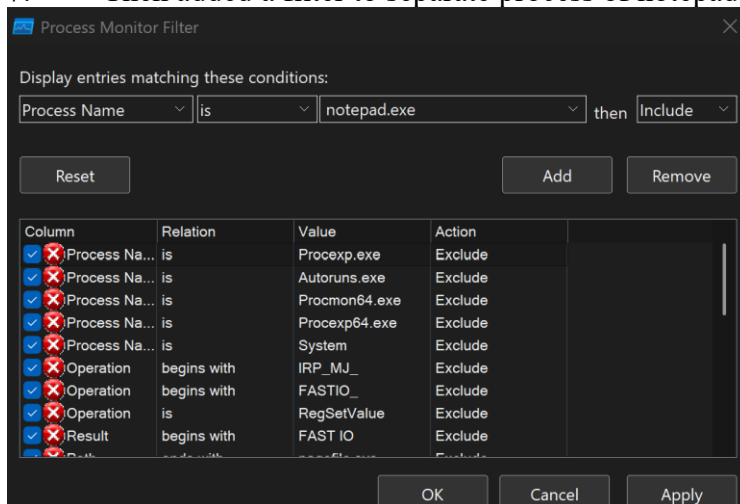
5. Opened Notepad and entered text into Notepad and save the file.



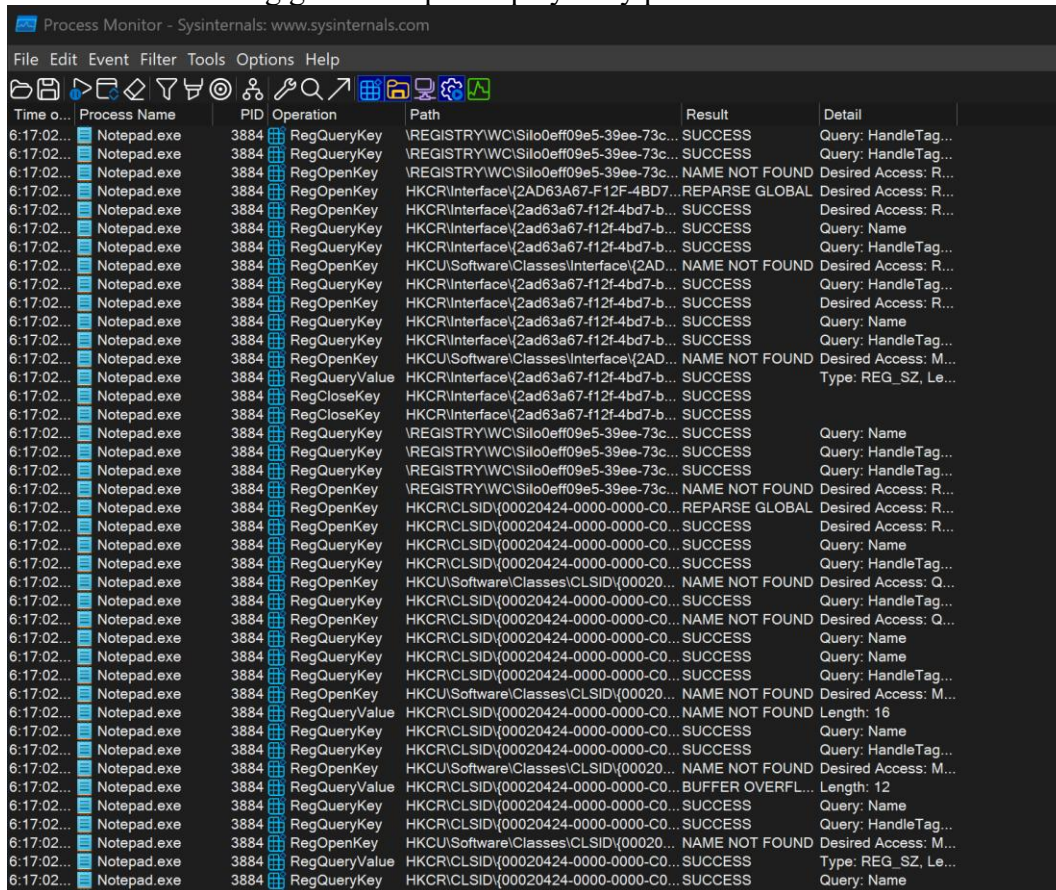
6. Tried filtering by Show Registry Activity filter on promon.



7. Then added a filter to separate process of notepad.



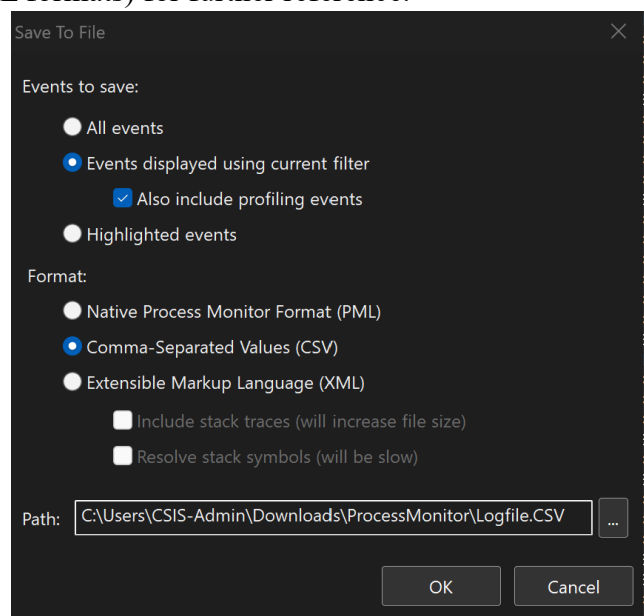
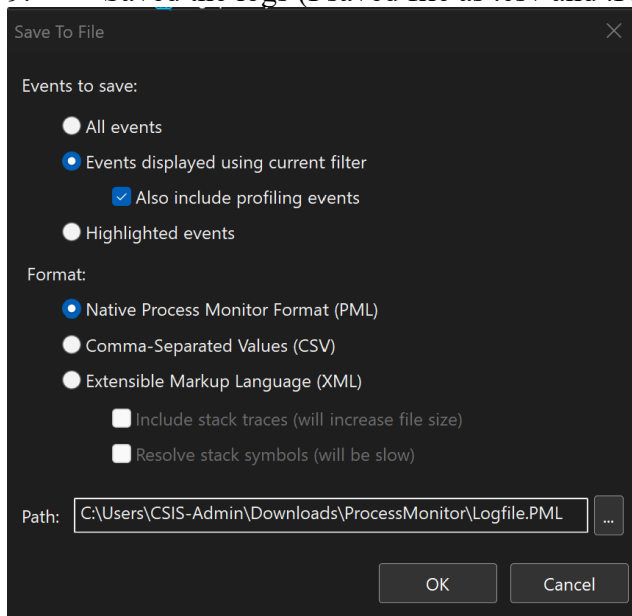
8. After filtering get the output displayed by procmon.



Process Monitor - Sysinternals: www.sysinternals.com

Time	Process Name	PID	Operation	Path	Result	Detail
6:17:02...	Notepad.exe	3884	RegQueryKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegQueryKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	NAME NOT FOUND	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	REPARSE GLOBAL	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\Interface\2AD...	NAME NOT FOUND	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\Interface\2AD...	NAME NOT FOUND	Desired Access: M...
6:17:02...	Notepad.exe	3884	RegQueryValue	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	Type: REG_SZ, Le...
6:17:02...	Notepad.exe	3884	RegCloseKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	
6:17:02...	Notepad.exe	3884	RegCloseKey	HKCR\Interface\2ad63a67-f12f-4bd7-b...	SUCCESS	
6:17:02...	Notepad.exe	3884	RegQueryKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegQueryKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	\REGISTRY\WC\Silo0eff09e5-39ee-73c...	NAME NOT FOUND	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\CLSID\{00020424-0000-0000-C0...	REPARSE GLOBAL	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Desired Access: R...
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\CLSID\{00020...	NAME NOT FOUND	Desired Access: Q...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\CLSID\{00020424-0000-0000-C0...	NAME NOT FOUND	Desired Access: Q...
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\CLSID\{00020...	NAME NOT FOUND	Desired Access: M...
6:17:02...	Notepad.exe	3884	RegQueryValue	HKCR\CLSID\{00020424-0000-0000-C0...	NAME NOT FOUND	Length: 16
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\CLSID\{00020...	NAME NOT FOUND	Desired Access: M...
6:17:02...	Notepad.exe	3884	RegQueryValue	HKCR\CLSID\{00020424-0000-0000-C0...	BUFFER OVERFL...	Length: 12
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: HandleTag...
6:17:02...	Notepad.exe	3884	RegOpenKey	HKCU\Software\Classes\CLSID\{00020...	NAME NOT FOUND	Desired Access: M...
6:17:02...	Notepad.exe	3884	RegQueryValue	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Type: REG_SZ, Le...
6:17:02...	Notepad.exe	3884	RegQueryKey	HKCR\CLSID\{00020424-0000-0000-C0...	SUCCESS	Query: Name

9. Saved the logs (I saved file as .csv and .PML formats) for further reference.



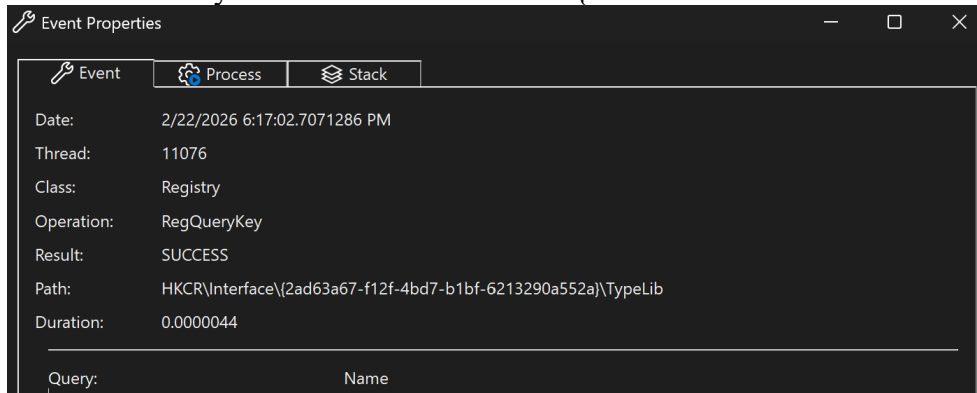
10. What registry keys did notepad write to?

- RegOpenKey
- RegQueryValue
- RegQueryKey
- RegEnumKey
- RegCloseKey

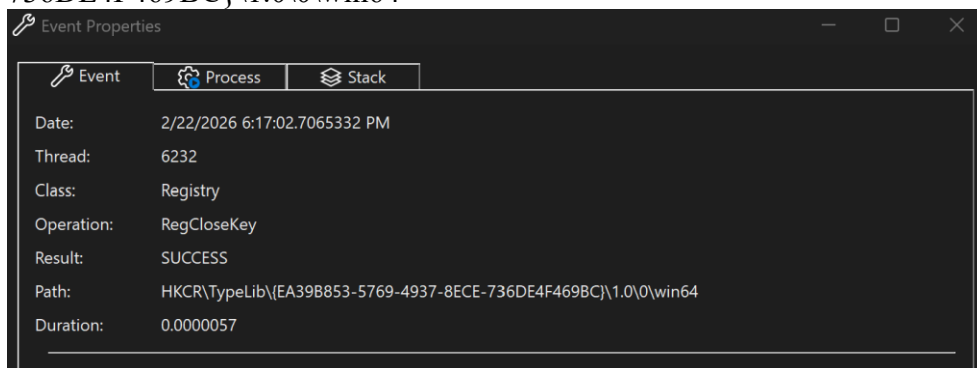
11. What keys did the program access?

ProcMon logs show that notepad.exe accessed multiple registry keys using RegOpenKey, RegQueryKey, RegQueryValue, and RegCloseKey operations.

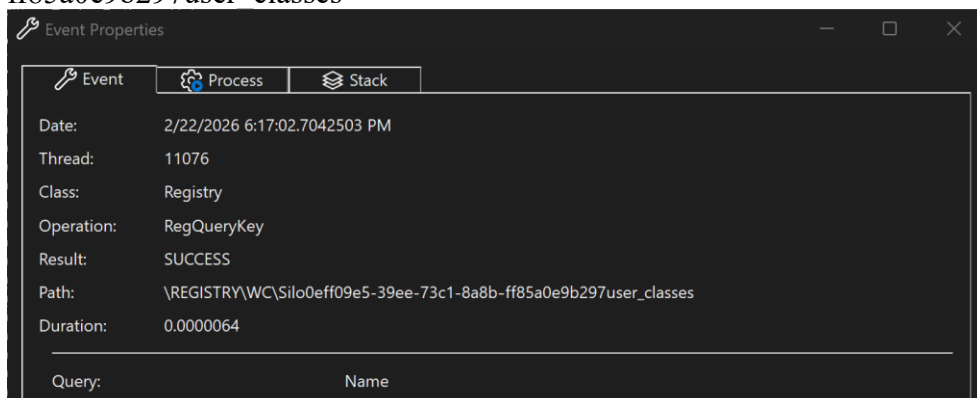
- Interface Key Access - HKCR\Interface\{2ad63a67-f12f-4bd7-b1bf-6213290a552a}\TypeLib



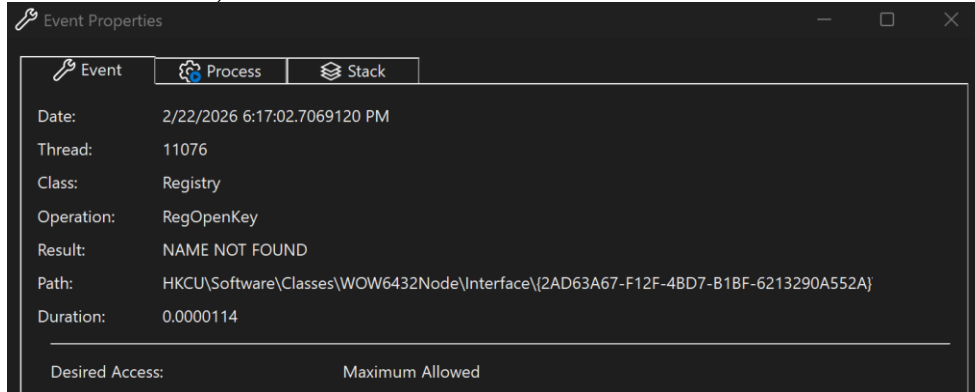
- Type Library Key Access - HKCR\TypeLib\{EA39B853-5769-4937-8ECE-736DE4F469BC}\1.0\0\win64



- Native Registry Path Access - \REGISTRY\WC\Silo0eff09e5-39ee-73c1-8a8b-ff85a0e9b297\user_classes



- HKCU\Software\Classes\WOW6432Node\Interface\{2AD63A67-F12F-4BD7-B1BF-6213290A552A}\Forward

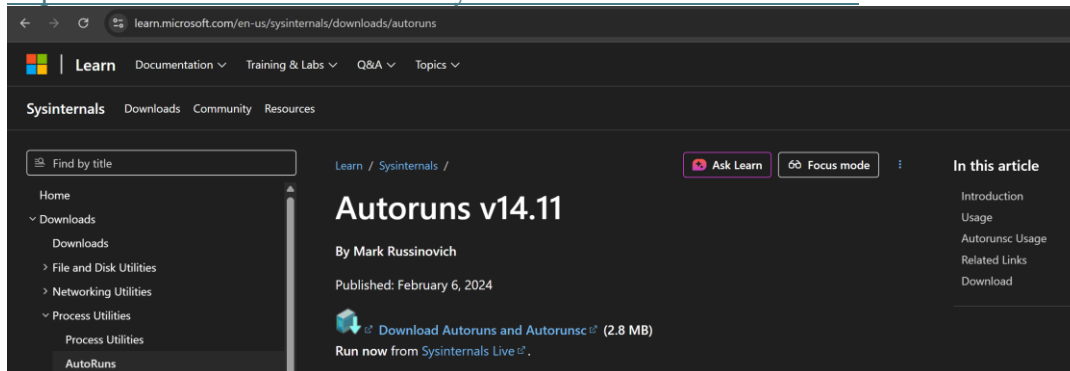


These entries include Interface entries under HKCR\Interface, TypeLib entries under HKCR\TypeLib, and user class entries under HKCU\Software\Classes, and native registry paths within the Windows registry structure.

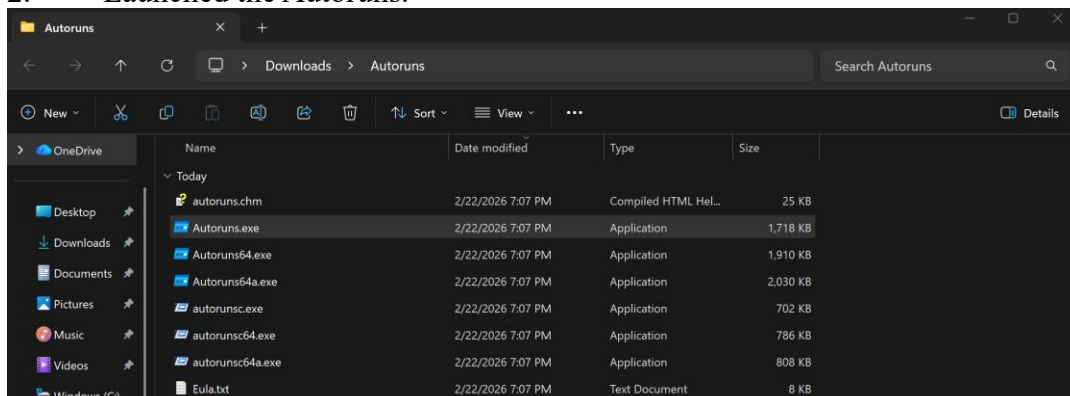
Part 2 - Autoruns

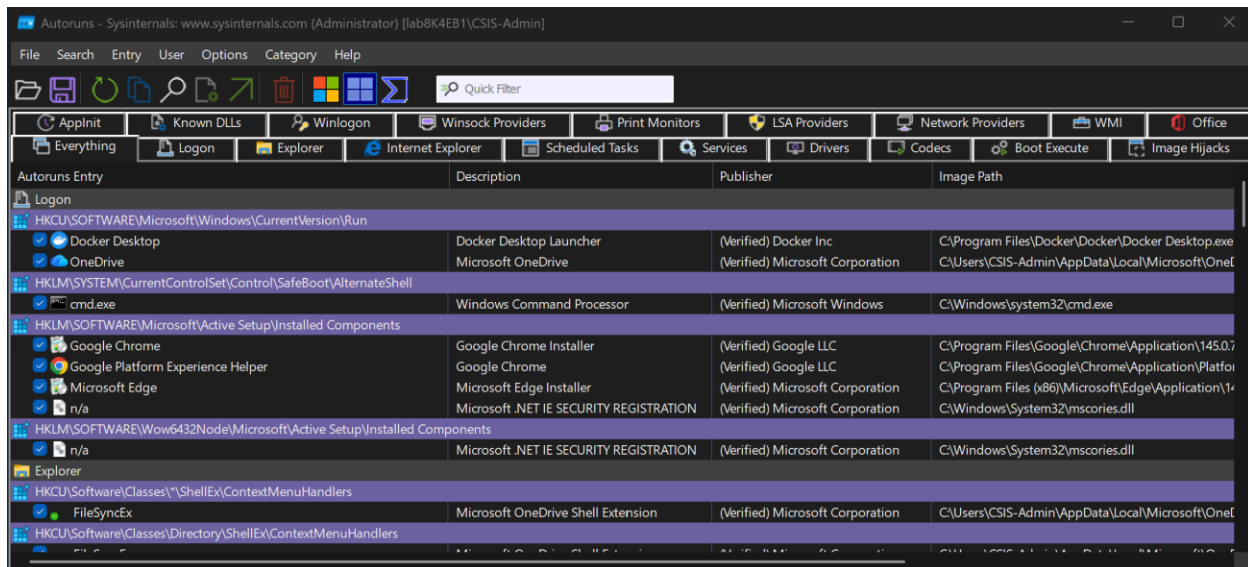
1. Downloaded the autorun using URL

<https://learn.microsoft.com/en-us/sysinternals/downloads/autoruns> and downloaded the Autoruns



2. Launched the Autoruns.





3. What is the Autoruns tool used for?

Autoruns is a Microsoft tool which shows what programs are configured to run during system startup or login, and when we start various built-in Windows applications like Internet Explorer, Explorer and media players.

By simply running *Autoruns*, we can get the currently configured auto-start applications and the full list of Registry and file system locations available for auto-start configuration. These details can be used in malware analysis and incident response to detect persistence mechanisms and unauthorized startup programs.

4. Describe two programs shown in the Autoruns application. I chose Docker Desktop and Google chrome from Autoruns application.

Autoruns - Sysinternals: www.sysinternals.com (Administrator) [lab8K4EB1\CSIS-Admin]				
File Search Entry User Options Category Help				
Quick Filter				
Winlogon Winsock Providers Print Monitors LSA Providers Network Providers WMI				
Everything Logon Explorer Internet Explorer Scheduled Tasks Services Drivers Codex Boot Execute Image Hijacks Applint				
Autoruns Entry	Description	Publisher	Image Path	Timestamp
Logon				
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run				Thu Jan 8 11:19:31 2026
Docker Desktop	Docker Desktop Launcher	(Verified) Docker Inc	C:\Program Files\Docker\Docker\Docker Desktop.exe	Thu Jan 8 11:18:31 2026
OneDrive	Microsoft OneDrive	(Verified) Microsoft Corporation	C:\Users\CSIS-Admin\AppData\Local\Microsoft\OneDrive\OneDrive.exe	Mon Feb 16 14:12:07 2026
HKLM\SYSTEM\CurrentControlSet\Control\SafeBoot\AlternateShell				Mon Apr 1 02:26:32 2024
cmd.exe	Windows Command Processor	(Verified) Microsoft Windows	C:\Windows\system32\cmd.exe	Wed Jan 7 14:08:32 2026
HKLM\SOFTWARE\Microsoft\Active Setup\Installed Components				Mon Feb 2 18:53:20 2026
Google Chrome	Google Chrome Installer	(Verified) Google LLC	C:\Program Files\Google\Chrome\Application\145.0.7632.109\Installer\chrome_installer.exe	Sun Feb 22 17:05:43 2026
Google Platform Experience Helper	Google Chrome	(Verified) Google LLC	C:\Program Files\Google\Chrome\Application\PlatformExperienceHelper.exe	Thu Jan 29 22:27:34 2026
Microsoft Edge	Microsoft Edge Installer	(Verified) Microsoft Corporation	C:\Program Files (x86)\Microsoft\Edge\Application\145.0.3800.70\Installer\MicrosoftEdgeSetup.exe	Sun Feb 22 17:05:17 2026
n/a	Microsoft .NET IE SECURITY REGISTRATION	(Verified) Microsoft Corporation	C:\Windows\System32\mscories.dll	Mon Apr 1 02:22:17 2024

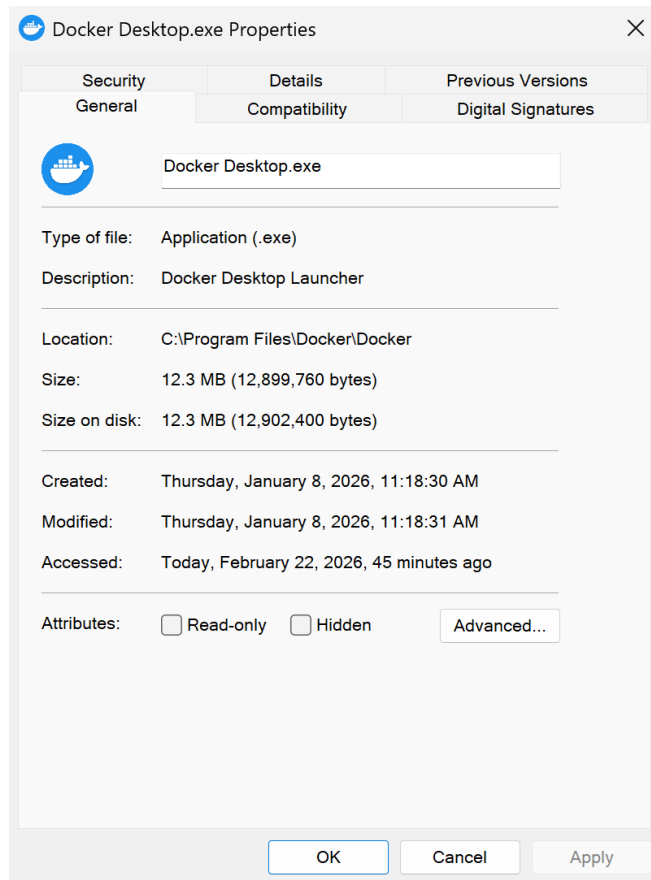
Docker Desktop

HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run				Thu Jan 8 11:19:31 2026
Docker Desktop	Docker Desktop Launcher	(Verified) Docker Inc	C:\Program Files\Docker\Docker\Docker Desktop.exe	Thu Jan 8 11:18:31 2026

- Registry Location- HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- Description - Docker Desktop Launcher
- Publisher - (Verified) Docker Inc

- Image Path - C:\Program Files\Docker\Docker\Docker Desktop.exe

- Timestamp - Thu Jan 8, 2026 11:18:31 AM



Docker Desktop is an application for Mac, Linux, or Windows environment that is used to build, share, and run containerized applications and microservices. It provides a straightforward GUI (Graphical User Interface) that can manage the containers, applications, and images directly from our machine. Its presence under the *Run* registry key indicates that it is configured to start automatically when the user logs in. This file indicates normal and trusted behavior because it is digitally signed by Docker Inc and located in a legitimate program directory.

Google Chrome

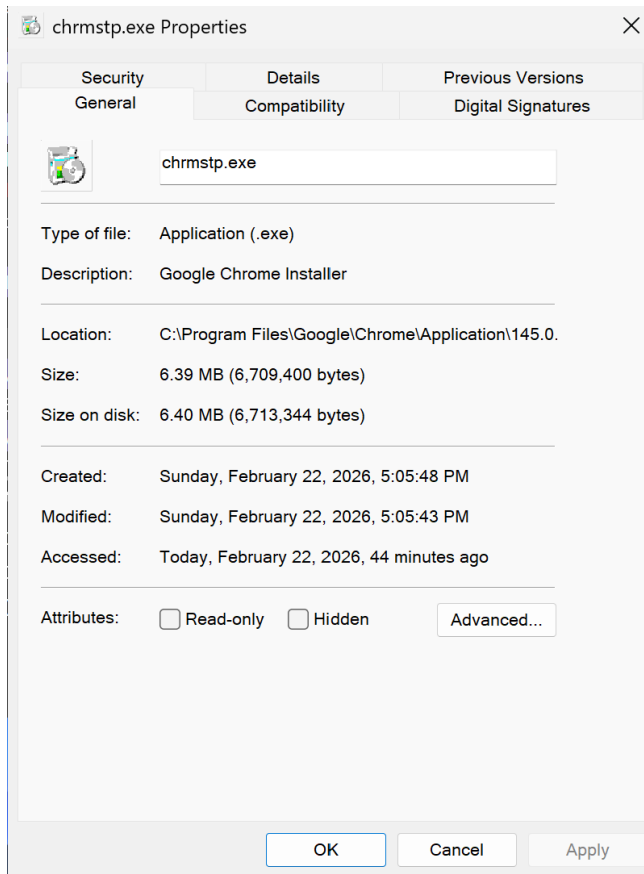


- Registry Location- HKLM\SOFTWARE\Microsoft\Active Setup\Installed Components

- Description - Google Chrome Installer

- Publisher - (Verified) Google LLC

- Image Path - C:\Program Files\Google\Chrome\Application\145.0.7632.109\Instaler\chromstp.exe
- Timestamp - Sun Feb 22, 2026 17:05:43 PM



chromstp.exe is the Google Chrome installer component. It is responsible for installing Google Chrome, managing updates, repairing installation files, and applying version upgrades. The *Active Setup* registry key is used by Windows to execute configuration tasks when a user logs in. Chrome uses this mechanism to finalize installation settings and apply updates automatically. This file indicates normal and trusted behavior because it is digitally signed by Google LLC and located in a legitimate program directory.

Conclusion

This lab helps to understand how legitimate Windows applications interact with the operating system and how to identify startup persistence mechanisms using Sysinternal tools. Registry activities associated with notepad.exe were captured using process monitor (procmon) and monitored the details of multiple registry keys, including Interface and TypeLib entries, through operations such as RegOpenKey and RegQueryValue. These details show how even simple applications perform numerous background registry queries during execution.

Startup entries for applications such as Docker Desktop and Google Chrome were identified using Autoruns. These programs were configured to execute at logon through registry

locations such as the Run key and Active Setup components. Autoruns also helps to verify publisher information and file paths when evaluating startup entries.

Most of the malicious software rely on registry modifications and startup entries to maintain persistence. By using tools like procmon and Autoruns, analysts can establish a baseline for normal system behavior and compare it against suspicious changes in the system.

References

1. Yosifovich, P., Ionescu, A., Russinovich, M. E., & Solomon, D. A. (2017). *Windows internals, part 1: System architecture, processes, threads, memory management, and more* (7th ed.). Microsoft Press.
2. Sikorski, M., & Honig, A. (2012). *Practical malware analysis: The hands-on guide to dissecting malicious software*. No Starch Press.
3. Fox, N. (2025, November 18). *How to use Autoruns to detect and remove malware on Windows*. Varonis. <https://www.varonis.com/blog/how-to-use-autoruns>
4. Russinovich, M. (2024, February 6). *Autoruns v14.11*. Microsoft Learn. <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>
5. Russinovich, M. (2024, June 20). *Process Monitor v4.01*. Microsoft Learn. <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>
6. Docker, Inc. (n.d.). *Docker Desktop*. Docker Docs. <https://docs.docker.com/desktop/>